CLEVER: A Portuguese Dataset for Claim Detection and Verification in Election News

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Synthetic Verification Question Generation for ClaimPT: Methodology and Evaluation

Supplementary documentation for Section 6 of the paper

To evaluate the potential of the CLEVER dataset, we formulate automatic verification question generation as a text-to-text generation task. Given a claim span extracted from a news article, its surrounding context, and the article's publication date, the model is tasked with generating one or more factual questions that capture the information needed to verify the claim.

To investigate the impact of incorporating additional training data, we also draw on the ClaimPT corpus (Campos et al., 2025[1], 2026[2]), which contains 522 claims. For these claims, 1,073 synthetic verification questions were generated, resulting in the Silver1073 dataset.

This document provides a detailed account of the methodology used to generate synthetic verification questions for the claims in ClaimPT, as well as the procedures employed to assess the quality of the generated questions.

1. Synthetic Verification Question Generation for ClaimPT

To expand the amount of training data available for verification question generation, an additional set of questions was automatically generated for ClaimPT. This process aimed to complement the manually annotated verification questions available in CLEVER by using the previously annotated claim spans from ClaimPT as the basis for synthetic question generation.

The questions were generated using Gemini 3.5 Flash, employed as a teacher model. A dedicated prompt was developed based on the question-formulation guidelines used in CLEVER. The prompt included detailed instructions specifying how verification questions should be formulated, together with six manually annotated examples from the CLEVER training set. These examples illustrated how the annotation criteria could be applied to different claim structures and factual information needs.

For each ClaimPT claim span, the model received the exact claim span, its surrounding textual context, the publication date of the source article, and the available annotations concerning the claim object, claimer, and temporal information associated with the claim. The claim object served

[1] Campos, R., Sequeira, R., Nerea, S., Cantante, I., Folques, D., Cunha, F., Canavilhas, J., Branco, A., Jorge, A., Nunes, S., Guimarães, N., & Silvano, P. (2025). ClaimPT: A dataset for claim detection and fact-checking [Data set]. INESC TEC. https://doi.org/10.25747/JY10-E413

[2] Campos, R., Sequeira, R., Nerea, S., Cantante, I., Folques, D., Cunha, L. F., . . . Silvano, P. (2026). ClaimPT: A Portuguese dataset of annotated claims in news articles. In R. Campos, A. Jatowt, Y. Lan, M. Aliannejadi, C. Bauer, S. MacAvaney, . . . S. Verberne (Eds.), Advances in information retrieval: 48th European Conference on Information Retrieval, ECIR 2026, proceedings (Lecture Notes in Computer Science, Vol. 16486, pp. 544–560). Berlin, Germany: Springer. https://doi.org/10.1007/978-3-032-21321-1_58

as the primary focus of question generation, while the remaining information was used to interpret the claim, resolve ambiguous references, identify the entities involved, and establish the appropriate temporal context.

The generation instructions required questions to focus strictly on the factual content expressed in the claim span, rather than on peripheral information from the source article. Questions were expected to be factual, neutral, clear, self-contained, and suitable for retrieving documentary evidence, without presupposing whether the claim was true or false. The model was also instructed to avoid redundant questions, replace ambiguous pronouns with identifiable entities, and incorporate temporal or geographical references whenever necessary for unambiguous verification. Additional instructions addressed claims involving comparisons, references to studies or reports, and factual information whose verification depended on the identity of the claimer.

Following the CLEVER annotation guidelines, the model was encouraged, whenever appropriate, to combine a general verification question, addressing the broader factual subject of the claim object, with more specific questions targeting the concrete information needed for verification, such as dates, quantities, entities, or events. Rather than enforcing a fixed number of questions, the model determined how many were needed according to the distinct verifiable subfacts identified in each claim span. This approach was intended to cover the relevant factual information while avoiding unnecessary paraphrases or multiple questions addressing the same fact.

Question generation was performed with a decoding temperature of 0.0, and the model was instructed to return only valid JSON in the format {"questions": ["Question 1?", "Question 2?"]}, without additional explanatory text. The resulting questions were formulated in European Portuguese and associated with their corresponding ClaimPT claim spans.

2. Manual Quality-Control Assessment

A manual quality-control assessment was conducted to evaluate the automatically generated questions before their experimental use. A simple random sample of 50 claim spans, comprising 104 synthetic verification questions, was selected using a fixed random seed.

The assessment examined two complementary dimensions: the general quality of individual questions and their adherence to the CLEVER question-formulation guidelines. This distinction was necessary because a question could be relevant, factually faithful, and clearly formulated while still failing to follow specific aspects of the annotation protocol, such as maintaining the intended relationship between general and specific questions.

The evaluation comprised eight criteria, grouped into these two dimensions. Depending on the criterion, compliance was assessed either for individual questions or for complete sets of questions associated with a claim span.

2.1. General Question Quality

Four criteria were used to assess the intrinsic quality of the generated questions.

Relevance. This criterion assessed whether a question directly addressed the factual content of its corresponding claim span. Questions focusing on peripheral information from the source article, rather than on the fact being verified, were considered non-compliant. A total of 100 out of 104 questions (96.2%) satisfied this criterion.

Faithfulness. This criterion assessed whether the question remained consistent with the claim span and its accompanying context. In particular, it checked whether the model introduced unsupported factual information, such as entities, numbers, dates, or other details not justified by the available input. 97 out of 104 questions (93.3%) satisfied this criterion. Faithfulness had the lowest compliance rate among the four general quality criteria.

Clarity and Searchability. This criterion evaluated whether each question was understandable independently of the original article and sufficiently explicit to support evidence retrieval. Questions were expected to resolve ambiguous pronouns and references by using identifiable entities, places, or other appropriate expressions. 100 out of 104 questions (96.2%) satisfied this criterion.

Temporal and Local Anchoring. This criterion assessed whether the questions included the relevant temporal or geographical information whenever necessary to identify the factual situation unambiguously. Such information could include dates, time periods, regions, or specific locations. Questions were not required to include these elements when they were unnecessary for verification. 103 out of 104 questions (99.0%) satisfied this criterion.

The complete results for general question quality are summarized below.

Criterion	**Compliant questions**	**Compliance rate**
Relevance	100/104	96.2%
Faithfulness	97/104	93.3%
Clarity and Searchability	100/104	96.2%
Temporal and Local Anchoring	103/104	99.0%

Overall, the results indicate that most questions in the evaluated sample were relevant to their corresponding claim spans, faithful to the supplied information, understandable in isolation, and appropriately contextualized for evidence retrieval.

2.2. Adherence to the CLEVER Question-Formulation Guidelines

Four additional criteria assessed whether the synthetic questions reproduced the structural and formulation principles followed during the manual annotation of CLEVER.

General-Specific Structure. This criterion evaluated the relationship between the questions generated for the same claim span. Whenever the claim allowed a meaningful distinction, the question set was expected to include a general question addressing the central factual subject and a more specific question targeting additional information required for verification. When such a distinction was not meaningful, the questions were expected to remain focused on the central fact without introducing redundant paraphrases. The criterion was evaluated at the claim-span level, considering the complete set of generated questions. 34 out of 50 claim spans (68.0%) satisfied this criterion.

Claim-Object Adherence. This criterion assessed whether each question remained centred on the claim object associated with its claim span. The claim object identifies the principal element or factual content that the question must address. Questions that shifted their focus towards other elements of the source article or unrelated aspects of the claim were considered non-compliant. 92 out of 104 questions (88.5%) satisfied this criterion.

Neutrality and Factual Focus. This criterion evaluated whether the generated questions sought factual information without presupposing the truth or falsity of the claim. Questions were expected to avoid subjective or leading formulations and to remain focused on the alleged fact rather than on the opinions, intentions, or credibility of the claimer. This requirement applied regardless of whether the original claim was annotated with an Affirm or Refute stance. 103 out of 104 questions (99.0%) satisfied this criterion.

Special-Case Compliance. This criterion assessed whether the generated questions correctly handled specific situations requiring additional information for verification. Three types of situations were considered:

- **Claimer-dependent facts:** when the validity of a factual statement depended on who made it, the relevant claimer had to be identified in the question.
- **Comparative claims:** when a claim established a comparison, the questions had to cover the factual information necessary to verify both points of comparison.
- **References to external documents:** when a claim referred to a study, report, opinion poll, or another identifiable document, the questions had to allow that source to be located.

These requirements were assessed only when applicable. At least one of the three conditions occurred in 30 of the 50 evaluated claim spans. When multiple conditions applied to the same claim span, all applicable requirements had to be satisfied for that claim span to be considered compliant. 26 out of these 30 claim spans (86.7%) satisfied this criterion.

The results for adherence to the CLEVER guidelines are summarized below.

Criterion	**Compliant units**	**Compliance rate**	**Evaluation unit**
General-Specific Structure	34/50	68.0%	Claim span
Claim-Object Adherence	92/104	88.5%	Question
Neutrality and Factual Focus	103/104	99.0%	Question
Special-Case Compliance	26/30	86.7%	Claim span with applicable conditions

These results show that the synthetic questions generally preserved the intended focus on the claim object, maintained a neutral factual formulation, and handled most applicable special cases appropriately.

However, General-Specific Structure achieved the lowest compliance rate (68.0%), indicating that reproducing the intended relationship between general and specific questions was the most challenging aspect of the annotation guidelines for the teacher model.

The manual assessment was conducted by the researcher responsible for the synthetic question generation process. It therefore served as a quality-control procedure rather than an independent annotation exercise. Furthermore, only a randomly selected sample was evaluated; the complete collection was not exhaustively validated by human annotators.

Consequently, the automatically generated questions are classified as silver-standard annotations, whereas the manually formulated CLEVER questions constitute gold-standard references.

The quality-control results indicate that the synthetic questions generally follow the intended annotation principles, but they do not guarantee that every generated question is correct or fully compliant with the guidelines. In particular, the observed differences in compliance across criteria highlight the importance of checking factual faithfulness, adherence to the claim object, and the relationship between general and specific questions when reviewing automatically generated annotations.

3. PROMPT

És um anotador especializado em produzir perguntas para verificação de claims em português europeu.

Recebes UM claim span de cada vez e devolves perguntas que servem para verificar a veracidade desse span.

Elementos disponíveis:

- Claim span: segmento textual exato cuja veracidade tem de ser validada.
- Claim object(s): elemento, entidade, valor ou número focal da alegação dentro do span.
- Claimer(s): quem profere a alegação.
- Time / Publication time: informação temporal usada para anchoring temporal.
- Contexto: artigo usado apenas para desambiguar pronomes, entidades e referências.

Princípio central:

- O objetivo é verificar a veracidade do CLAIM SPAN.
- Todas as perguntas devem incidir diretamente sobre o claim span e o seu conteúdo factual.
- Contexto, data, claimer e claim object são apenas auxiliares para interpretar, desambiguar e situar temporalmente a claim.
- Não gerar perguntas sobre informação periférica do artigo.

Critérios para formulação de perguntas para verificação de claims:

1. Formulação das questões por segmento da claim

- As perguntas devem estar centradas obrigatoriamente no elemento claim span.
- Cada pergunta deve estar estritamente vinculada ao claim object correspondente ao seu span.

2. Quantas perguntas gerar (cardinalidade)

- O número de perguntas segue o número de subfactos distintos e verificáveis do claim span - tipicamente entre 1 e 4.
- Se o span contém um único facto indivisível, gerar UMA pergunta. Não a duplicar com uma paráfrase.
- Se contém dois ou mais subfactos distintos (números, datas, entidades, comparações, ações), gerar uma pergunta por subfacto, até 4.
- Quando desdobrável, é útil combinar uma pergunta de nível geral (temática ampla do claim object) com perguntas de nível específico (números, datas, nomes, valores, instituições).
- NÃO gerar paráfrases do mesmo subfacto.
- Utilizar os links de identity para substituir pronomes, como "este sistema", "isso", "elas" ou "lá", pelos nomes próprios ou comuns das entidades, garantindo clareza para a busca.

3. Critérios adicionais

- Neutralidade: evitar termos subjetivos ou que induzam confirmação/negação.
- Independentemente de o stance ser Affirm ou Refute, a pergunta deve ser imparcial e focar-se no facto alegado.
- Anchoring temporal: sempre que o elemento Time estiver presente, ou puder ser inferido via identity com a publication time, incluir a baliza temporal.
- Tópicos locais: mencionar explicitamente a região ou entidade territorial referida.
- Integrar o claimer apenas quando a validade do facto depende da autoridade de quem o profere.
- Se a alegação referir estudos ou relatórios, estruturar a pergunta para localizar o documento específico.
- Em claims comparativas, desdobrar a verificação para obter os valores de ambos os pontos de comparação.

Etapas para a formulação:

1. Delimitar o claim span e garantir que tem significado completo.
2. Isolar o claim object - o termo mais específico dentro do claim span.
3. Identificar os subfactos distintos e verificáveis do span; o número de perguntas segue esses subfactos.
4. Usar claimer, time, identity e contexto apenas para desambiguar, resolver pronomes e situar temporalmente.

5. Escrever uma pergunta pesquisável por cada subfacto.

Formato de saída:

Devolve APENAS JSON válido, sem texto adicional, no formato:

{"questions": ["Pergunta 1?", "Pergunta 2?"]}

CLAIM SPAN: a criminalidade violenta aumentou

CLAIM OBJECT(S):

- criminalidade violenta

CLAIMER(S):

- Pedro Pinto

- O deputado do CDS-PP João Almeida

DATA DE PUBLICAÇÃO: 2025-04-16

SAÍDA: {"questions": ["A criminalidade violenta aumentou muito em comparação com outros anos em Portugal, à data de 16 de abril de 2025?"]}

CLAIM SPAN: Não há um exemplo onde tenha funcionado aquilo

CLAIM OBJECT(S):

- tenha funcionado aquilo

CLAIMER(S):

- O líder da IL

- Rui Rocha

DATA DE PUBLICAÇÃO: 2025-05-05

SAÍDA: {"questions": ["O teto às rendas funcionou alguma vez em Portugal até ao dia 5 de maio de 2025?", "Como funcionou o teto às rendas na Europa em maio de 2025?"]}

CLAIM SPAN: Cavaco Silva, 13 dias antes de o BES cair, disse que estava tudo bem

CLAIM OBJECT(S):

- disse que estava tudo bem

CLAIMER(S):

- O líder do PS

- Pedro Nuno Santos

DATA DE PUBLICAÇÃO: 2025-05-05

SAÍDA: {"questions": ["Cavaco Silva disse que estava tudo bem, 13 dias antes de o BES cair?", "O que disse Cavaco Silva, 13 dias antes de o BES cair?"]}

CLAIM SPAN: a Cristina Rodrigues é acusada pelo seu antigo partido de ter levado a cabo atos menos próprios

CLAIM OBJECT(S):

- levado a cabo atos menos próprios

CLAIMER(S):

- O líder do Chega

- André Ventura

DATA DE PUBLICAÇÃO: 2025-05-06

SAÍDA: {"questions": ["Em maio de 2025, Cristina Rodrigues era acusada de ter atos menos próprios?", "A acusação de Cristina Rodrigues de ter atos menos próprios foi feita pelo seu antigo partido, PAN?", "Quando foi feita a acusação sobre Cristina Rodrigues ter atos menos próprios?"]}

CLAIM SPAN: Portugal está muito aquém do cumprimento desse objetivo

CLAIM OBJECT(S):

- cumprimento desse objetivo

CLAIMER(S):

- A líder do PAN

- Inês de Sousa Real

DATA DE PUBLICAÇÃO: 2025-05-07

SAÍDA: {"questions": ["Em 2025, a erradicação das barracas era um objetivo longe de cumprir?", "Em 2025, o número de barracas estava a aumentar?", "Quantas barracas já foram erradicadas, em Portugal?"]}

CLAIM SPAN: Hoje, em muitos sítios de Portugal, também aqui na região de Setúbal, é muito difícil para muitas grávidas terem acesso a coisas tão simples, tão evidentes, como ecografias

CLAIM OBJECT(S):

- acesso

CLAIMER(S):

- Rui Rocha

DATA DE PUBLICAÇÃO: 2025-05-04

SAÍDA: {"questions": ["A quatro de maio de 2025, é difícil na região de Setúbal as grávidas terem acesso a ecografias?", "A quatro de maio de 2025, é difícil na região de Setúbal as grávidas terem acesso a vários cuidados de saúde?", "A quatro de maio de 2025, é difícil a nível nacional as

grávidas terem acesso a ecografias?", "A quatro de maio de 2025, é difícil a nível nacional as grávidas terem acesso a vários cuidados de saúde?"]}

Tarefa: gerar perguntas google-query-like que verifiquem a veracidade do CLAIM SPAN abaixo, usando o claim object como foco principal e os restantes elementos apenas como apoio para interpretar a alegação.

Não gerar perguntas para informação periférica do contexto. Gerar perguntas apenas para o claim span.

CLAIM SPAN:
{claim_span}
CONTEXTO:
{context}
DATA DE PUBLICAÇÃO:
{publication_time}
TIME ASSOCIADO À CLAIM:
{claim_time}
CLAIMER(S):
{claimers}
CLAIM OBJECT(S):
{claim_objects}

Instruções finais:
- Gera uma pergunta por cada subfacto distinto e verificável do span (tipicamente 1 a 4).
- Se o span tem um só subfacto, gera apenas 1 pergunta. Não gerar paráfrases do mesmo subfacto.
- Aplica anchoring temporal sempre que pertinente.
- Em tópicos locais, menciona explicitamente a região ou entidade territorial.
- Substitui pronomes pelos nomes próprios e comuns das entidades.
- Devolve apenas JSON válido, no formato:
{"questions": ["Pergunta 1?", "Pergunta 2?"]}

English translation

You are an expert annotator specialised in generating questions for claim verification in English.
You receive ONE claim span at a time and return questions designed to verify the factual accuracy of that span.

Available Elements

- Claim span: The exact textual segment whose factual accuracy must be verified.
- Claim object(s): The focal element, entity, value, or number of the claim within the span.
- Claimer(s): The person or entity making the claim.
- Time / Publication time: Temporal information used for temporal anchoring.
- Context: The article, used exclusively to disambiguate pronouns, entities, and references.

Core Principle

- The objective is to verify the factual accuracy of the CLAIM SPAN.
- All questions must directly address the claim span and its factual content.
- Context, date, claimer, and claim object are only auxiliary elements used to interpret, disambiguate, and temporally situate the claim.
- Do not generate questions about peripheral information from the article.

Criteria for Formulating Claim Verification Questions

1. Question formulation based on the claim span

- Questions must be centred exclusively on the claim span.
- Each question must be strictly linked to the claim object corresponding to its span.

2. Number of questions to generate (cardinality)

- The number of questions must correspond to the number of distinct, verifiable subfacts contained in the claim span, typically between 1 and 4.
- If the span contains a single indivisible fact, generate ONE question. Do not duplicate it through paraphrasing.
- If the span contains two or more distinct subfacts (numbers, dates, entities, comparisons, actions), generate one question per subfact, up to a maximum of 4.
- When the claim can be decomposed, it may be useful to combine a general-level question (addressing the broader topic of the claim object) with specific-level questions (numbers, dates, names, values, institutions).
- DO NOT generate paraphrases of the same subfact.
- Use identity links to replace pronouns and anaphoric expressions, such as "this system", "that", "they", or "there", with the proper names or common nouns of the corresponding entities, ensuring clarity for information retrieval.

3. Additional criteria

- Neutrality: Avoid subjective or leading formulations that suggest confirmation or refutation.

- Stance independence: Regardless of whether the stance is Affirm or Refute, the question must remain impartial and focus on the alleged fact.
- Temporal anchoring: Whenever a Time element is present, or can be inferred through identity links and the publication time, include the relevant temporal reference.
- Local topics: Explicitly mention the geographical region or territorial entity referred to.
- Claimer relevance: Include the claimer only when the validity of the fact depends on the authority or identity of the person making the statement.
- Studies and reports: If the claim refers to studies or reports, formulate the question so that the specific document can be located.
- Comparative claims: Decompose the verification into questions that allow the values of both points of comparison to be retrieved.

Question Generation Steps

1. Identify the claim span and ensure that it conveys a complete meaning.
2. Isolate the claim object — the most specific term within the claim span.
3. Identify the distinct and verifiable subfacts within the span; the number of questions must correspond to these subfacts.
4. Use claimer, time, identity links, and context only to disambiguate, resolve pronouns, and establish temporal anchoring.
5. Write one search-oriented question for each distinct subfact.

Output Format

Return ONLY valid JSON, without any additional text, in the following format:

{"questions": ["Question 1?", "Question 2?"]}

CLAIM SPAN:

violent crime has increased

CLAIM OBJECT(S):

- violent crime

CLAIMER(S):

- Pedro Pinto
- CDS-PP MP João Almeida

PUBLICATION DATE: 2025-04-16

OUTPUT:

{"questions": ["Had violent crime increased significantly compared with previous years in Portugal as of April 16, 2025?"]}

CLAIM SPAN:

There is not a single example where that has worked

CLAIM OBJECT(S):

- that has worked

CLAIMER(S):

- The leader of IL
- Rui Rocha

PUBLICATION DATE: 2025-05-05

OUTPUT:

{"questions": ["Have rent caps ever worked in Portugal up to May 5, 2025?", "How did rent caps work in Europe in May 2025?"]}

CLAIM SPAN:

Cavaco Silva, 13 days before BES collapsed, said that everything was fine

CLAIM OBJECT(S):

- said that everything was fine

CLAIMER(S):

- The leader of PS
- Pedro Nuno Santos

PUBLICATION DATE: 2025-05-05

OUTPUT:

{"questions": ["Did Cavaco Silva say that everything was fine 13 days before BES collapsed?", "What did Cavaco Silva say 13 days before BES collapsed?"]}

CLAIM SPAN:

Cristina Rodrigues is accused by her former party of having engaged in inappropriate conduct

CLAIM OBJECT(S):

- having engaged in inappropriate conduct

CLAIMER(S):

- The leader of Chega
- André Ventura

PUBLICATION DATE: 2025-05-06

OUTPUT:

{"questions": ["In May 2025, was Cristina Rodrigues accused of inappropriate conduct?", "Was the accusation that Cristina Rodrigues had engaged in inappropriate conduct made by her former

party, PAN?", "When was the accusation concerning Cristina Rodrigues's alleged inappropriate conduct made?"]}

CLAIM SPAN:

Portugal is far from achieving that goal

CLAIM OBJECT(S):

- achieving that goal

CLAIMER(S):

- The leader of PAN
- Inês de Sousa Real

PUBLICATION DATE: 2025-05-07

OUTPUT:

{"questions": ["In 2025, was the goal of eliminating shacks far from being achieved?", "In 2025, was the number of shacks increasing?", "How many shacks had already been eliminated in Portugal?"]}

CLAIM SPAN:

Today, in many places in Portugal, including here in the Setúbal region, it is very difficult for many pregnant women to access things as simple and basic as ultrasound scans

CLAIM OBJECT(S):

- access

CLAIMER(S):

- Rui Rocha

PUBLICATION DATE: 2025-05-04

OUTPUT:

{"questions": ["On May 4, 2025, was it difficult for pregnant women in the Setúbal region to access ultrasound scans?", "On May 4, 2025, was it difficult for pregnant women in the Setúbal region to access various healthcare services?", "On May 4, 2025, was it difficult for pregnant women across Portugal to access ultrasound scans?", "On May 4, 2025, was it difficult for pregnant women nationwide to access various healthcare services?"]}

Task

Generate Google-query-like questions to verify the factual accuracy of the CLAIM SPAN below, using the claim object as the primary focus and the remaining elements only as supporting information for interpreting the claim.

Do not generate questions about peripheral information from the context. Generate questions only about the claim span.

CLAIM SPAN:

{claim_span}

CONTEXT:

{context}

PUBLICATION DATE:

{publication_time}

TIME ASSOCIATED WITH THE CLAIM:

{claim_time}

CLAIMER(S):

{claimers}

CLAIM OBJECT(S):

{claim_objects}

Final Instructions

- Generate one question for each distinct and verifiable subfact within the span (typically 1 to 4).
- If the span contains only one subfact, generate only 1 question. Do not generate paraphrases of the same subfact.
- Apply temporal anchoring whenever relevant.
- For local topics, explicitly mention the geographical region or territorial entity.
- Replace pronouns with the proper names or common nouns of the corresponding entities.
- Generate all questions in English.
- Return only valid JSON, in the following format:

{"questions": ["Question 1?", "Question 2?"]}